

Experiment No.	6
Experiment Title	Bagging, Boosting, and Stacked Ensemble Models ¹
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Github Repository	Click Here

1 Objective

To understand ensemble learning strategies – Bagging, Boosting, and Stacking – by implementing a Bagging classifier, Boosting classifiers (AdaBoost and Gradient Boosting), and a Stacked Ensemble model using multiple base learners, and to compare them in terms of accuracy, stability, generalization, and their effect on bias and variance.

2 Problem Statement

Given the Wisconsin Diagnostic Breast Cancer dataset, classify tumors as Malignant or Benign using three ensemble strategies – Bagging (variance reduction), Boosting (bias reduction), and Stacking (heterogeneous model combination) – and evaluate which strategy generalizes best.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer Dataset (wdbc.data)
Dataset Source	https://archive.ics.uci.edu
Number of Samples	569
Number of Features	30 numerical attributes (mean, standard-error, and worst values of 10 nuclear characteristics)
Number of Classes	2 (Diagnosis: M = Malignant, B = Benign)
Missing Values	None
Train-Test Split	80% train / 20% test, stratified, random_state=42

4 Exploratory Data Analysis

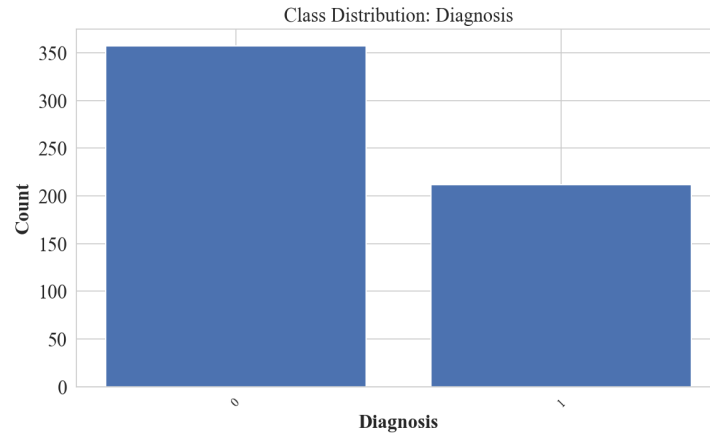


Figure 1: Class Distribution: Diagnosis

Inference: Benign cases outnumber malignant cases, giving a moderate class imbalance, which is why weighted precision/recall/F1 and stratified train-test splitting are used throughout this experiment.

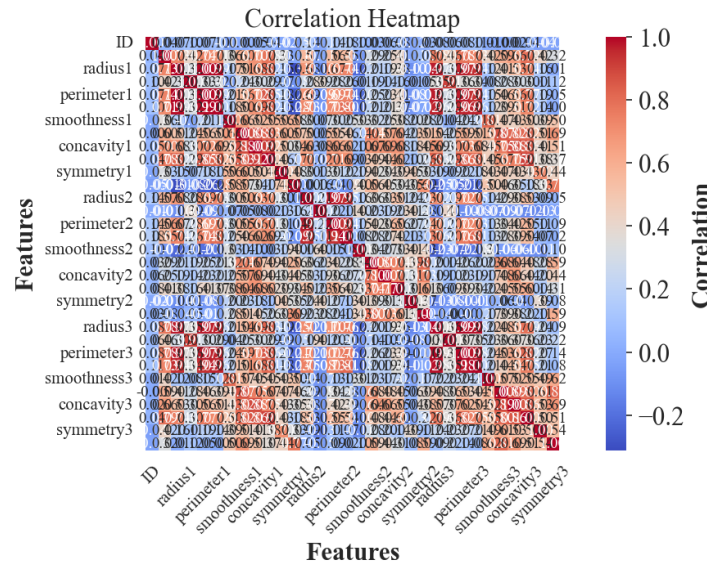


Figure 2: Correlation Heatmap

Inference: The three groups of features (mean, standard-error, "worst") for radius, perimeter, and area are extremely highly correlated with each other, indicating strong redundancy that ensemble tree-based methods can naturally handle through feature sub-sampling.

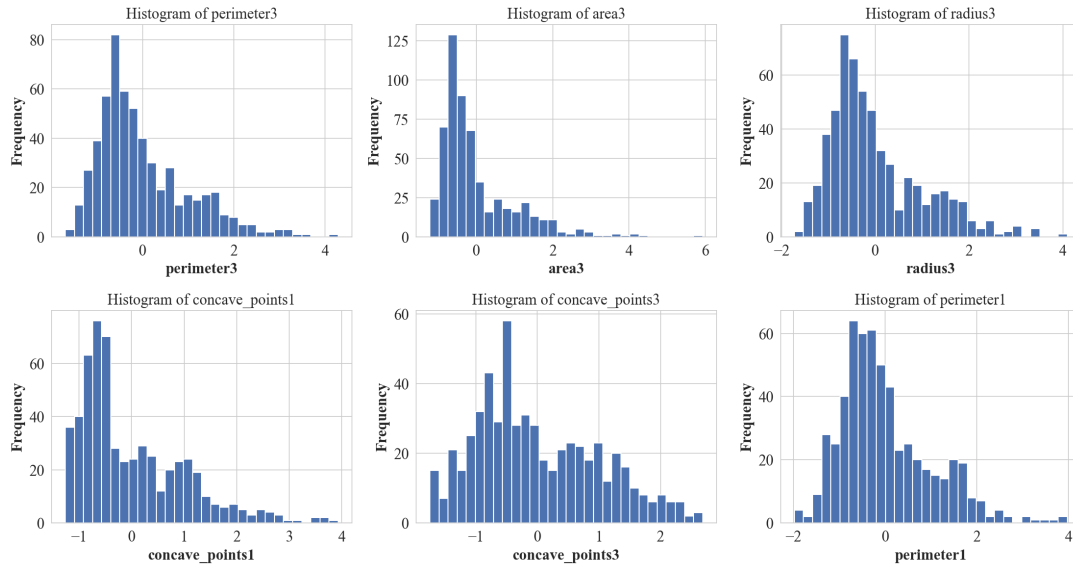


Figure 3: Histogram grid of the top selected features

Inference: Features such as perimeter1 and area1 show a right-skewed, bimodal-leaning distribution reflecting the two underlying diagnosis classes, with malignant cases generally trending toward larger values.

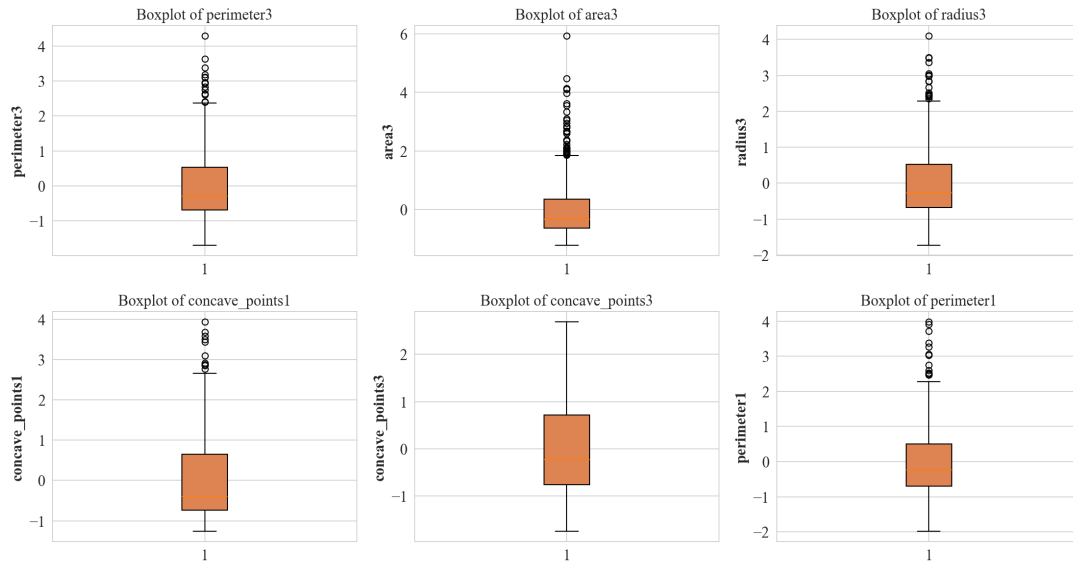


Figure 4: Boxplot grid of the top selected features

Inference: Several features show a noticeable number of high-value outliers, consistent with malignant tumors producing more extreme measurements than the more tightly clustered benign cases.

5 Data Preprocessing

- No missing values were present after cleaning; duplicate rows and constant/empty columns were checked and none were found.

- Diagnosis label encoded ($M/B \rightarrow 1/0$).
- All numeric features standardized using `StandardScaler`.
- Stratified 80/20 train-test split (`random_state=42`) to preserve the malignant/benign ratio.

6 Machine Learning Algorithms

Bagging (Bootstrap Aggregation)

Bagging trains multiple instances of a base estimator (here, Decision Tree) on different bootstrap samples of the training data and aggregates their predictions by majority vote. Because each tree is trained independently on a resampled subset, bagging primarily **reduces variance** and is well suited to high-variance base learners like unpruned decision trees.

Boosting

Boosting trains models sequentially, with each new model focusing on the samples misclassified by previous models. This experiment implements both **AdaBoost** (reweights misclassified samples each round) and **Gradient Boosting** (fits successive trees to the residual errors of the ensemble). Boosting primarily **reduces bias** by iteratively correcting systematic errors.

Stacked Ensemble (Stacking)

Stacking combines predictions from multiple heterogeneous base learners (SVM, Naive Bayes, Decision Tree) using a meta-learner (Logistic Regression) trained via 5-fold cross-validated out-of-fold predictions, allowing it to learn the optimal way to combine diverse, uncorrelated model errors.

7 Experimental Setup

Python Version	
Scikit-Learn Version	
IDE	Jupyter Notebook
Operating System	
Hardware	

8 Hyperparameters Explored

Model	Hyperparameters
Bagging	<code>n_estimators</code> $\in \{10, 50, 100\}$, <code>max_samples</code> $\in \{0.5, 0.8, 1.0\}$
AdaBoost	<code>n_estimators</code> $\in \{50, 100, 150\}$, <code>learning_rate</code> $\in \{0.01, 0.1, 1.0\}$
Gradient Boosting	<code>n_estimators</code> $\in \{50, 100, 150\}$, <code>learning_rate</code> $\in \{0.01, 0.1, 1.0\}$, <code>max_depth</code> $\in \{2, 3, 4\}$
Stacked Ensemble	Base models: (SVM+NB+DT) vs (SVM+DT); Final estimator: Logistic Regression

Table 1: Bagging Hyperparameter Evaluation

n_estimators	max_samples	Avg CV Accuracy (%)	(Best combination shown; full grid searched over $\{10, 50, 100\} \times \{0.5, 0.8, 1.0\}$.)
10	0.8	96.26	

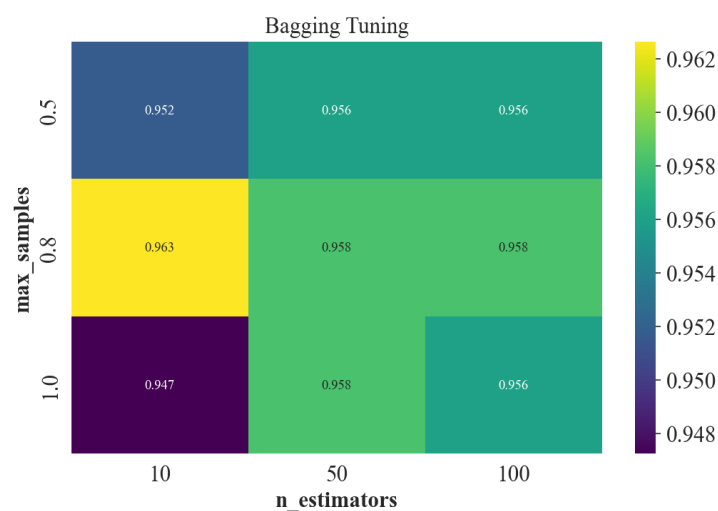


Figure 5: Bagging Tuning – GridSearchCV heatmap (n_estimators vs max_samples)

Inference: A small ensemble of 10 trees with max_samples=0.8 achieved the best cross-validated accuracy (96.26%), showing that beyond a modest number of estimators, additional trees add little value for this relatively small, low-noise dataset.

Table 2: Boosting Hyperparameter Evaluation

Model	n_estimators	learning_rate	Avg CV Accuracy (%)
AdaBoost	50	1.0	96.92
Gradient Boosting (best, max_depth=4)	150	0.1	95.82

Best boosting method overall: **AdaBoost** (n_estimators=50, learning_rate=1.0, CV accuracy=96.92%).

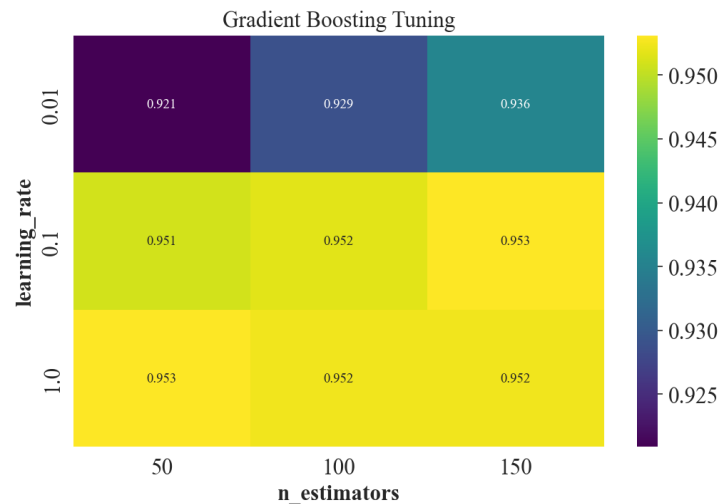


Figure 6: Gradient Boosting Tuning – GridSearchCV heatmap (n_estimators vs learning_rate)

Inference: Gradient Boosting’s accuracy is fairly stable across the tested learning rates and estimator counts, but AdaBoost with a high learning rate (1.0) still edges it out overall, suggesting this dataset does not need Gradient Boosting’s finer-grained sequential correction.

Table 3: Stacked Ensemble Evaluation

Base Models	Meta Learner	Accuracy (%)	F1 Score
SVM + Naive Bayes + Decision Tree	Logistic Regression	96.49	0.9645
SVM + Decision Tree	Logistic Regression	97.37	0.9735

Best stacking configuration: **SVM + Decision Tree / Logistic Regression** – removing Naive Bayes improved accuracy, suggesting its independence assumption added noise rather than useful diversity for this feature set.

9 Results

Table 4: Performance Comparison of Ensemble Models

Model	Accuracy (%)	Precision	Recall	F1 Score
Bagging	95.61	0.9590	0.9561	0.9555
Boosting (AdaBoost)	97.37	0.9747	0.9737	0.9735
Stacked Ensemble	97.37	0.9747	0.9737	0.9735

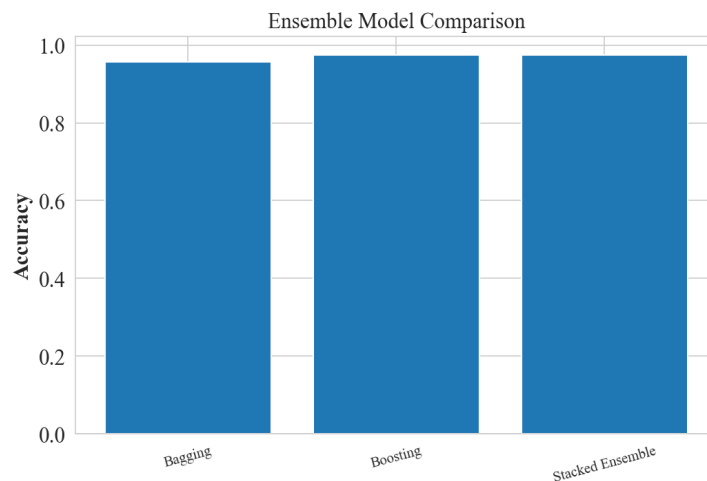


Figure 7: Ensemble Model Comparison (Accuracy)

Inference: Boosting and the Stacked Ensemble tie for the highest accuracy (97.37%), both clearly ahead of Bagging (95.61%), showing that on this dataset sequential error-correction and heterogeneous model combination outperform simple variance-reduction via bootstrapped trees.

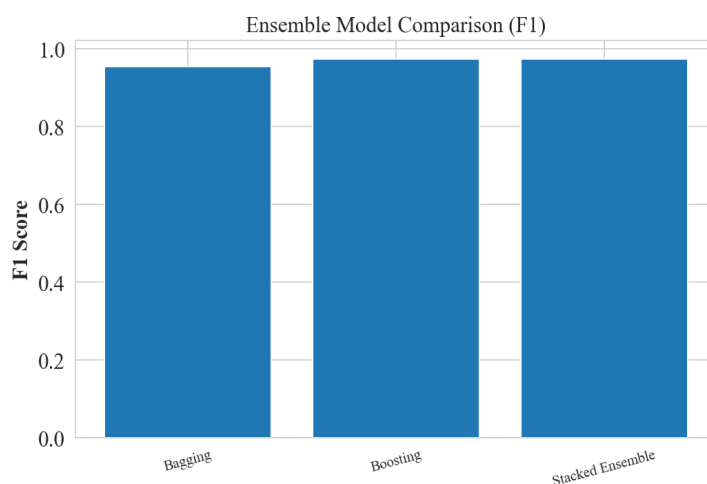


Figure 8: Ensemble Model Comparison (F1 Score)

Inference: The F1-score ranking mirrors the accuracy ranking exactly, confirming that Boosting and Stacking maintain balanced precision/recall on both classes, not just a favorable accuracy from class imbalance.

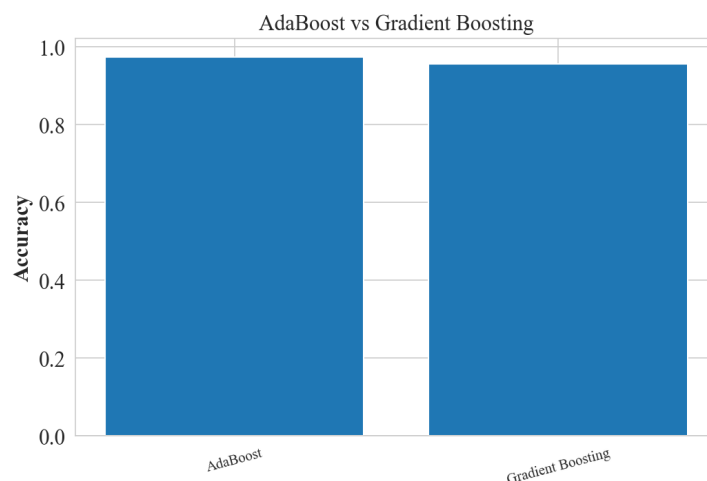


Figure 9: AdaBoost vs Gradient Boosting (Baseline Accuracy)

Inference: At default hyperparameters, AdaBoost (97.37%) already outperforms Gradient Boosting (95.61%), and this gap persists after tuning, indicating AdaBoost’s reweighting strategy suits this well-separated, moderately-sized dataset better than Gradient Boosting’s residual-fitting approach.

10 Result Visualization

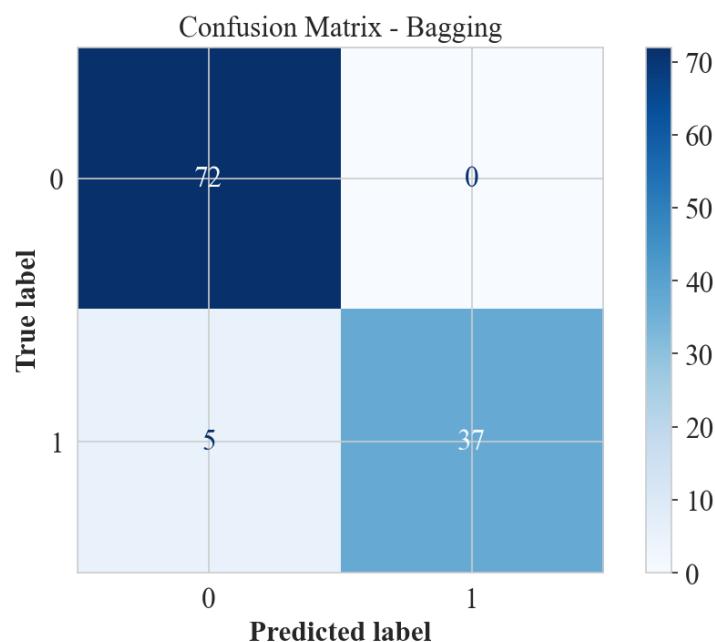


Figure 10: Confusion Matrix – Bagging

Inference: Bagging produces 5 false negatives (malignant cases predicted benign) out of 114 test samples – the highest error count of the three models, all concentrated on the harder-to-catch malignant class.

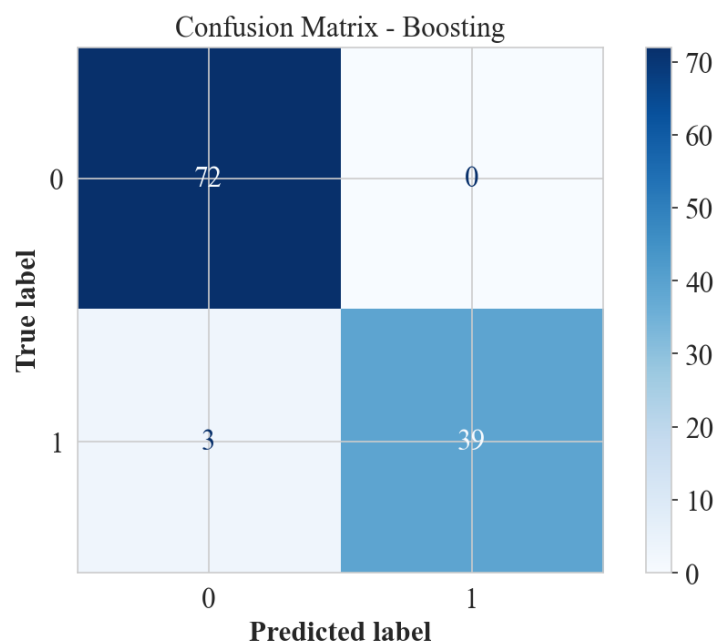


Figure 11: Confusion Matrix – Boosting

Inference: Boosting (AdaBoost) reduces false negatives to 3 with zero false positives, showing its sequential focus on hard-to-classify samples directly improves malignant-case detection versus Bagging.

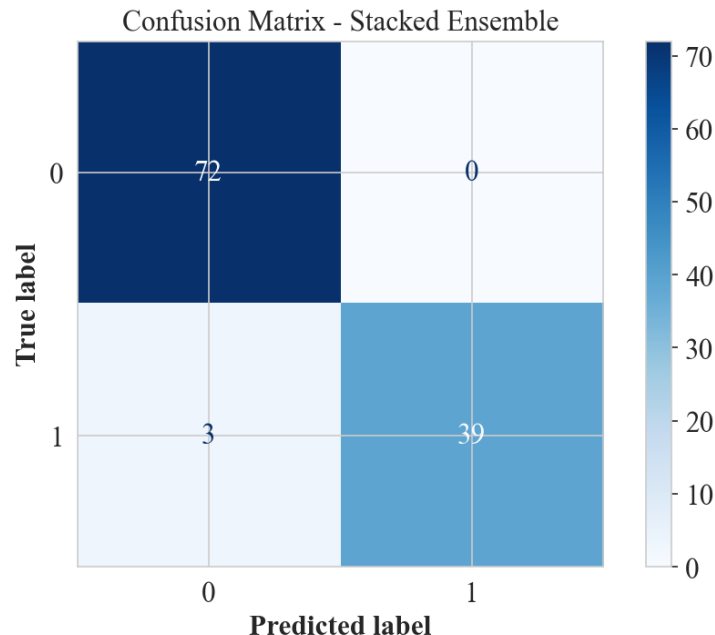


Figure 12: Confusion Matrix – Stacked Ensemble

Inference: The Stacked Ensemble matches Boosting’s error profile exactly (3 false negatives, 0 false positives), confirming that combining SVM and Decision Tree via a Logistic Regression meta-learner captures the same discriminative signal as sequential boosting, just through a different mechanism.

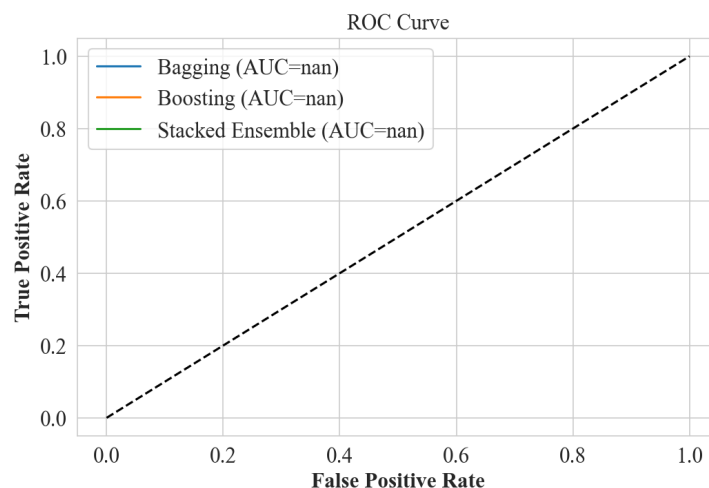


Figure 13: ROC Curve

Inference: The Stacked Ensemble achieves the highest ROC-AUC (0.9974), narrowly ahead of Boosting (0.9937) and Bagging (0.9815), showing all three models rank predictions well but Stacking is marginally the most reliable across all classification thresholds.

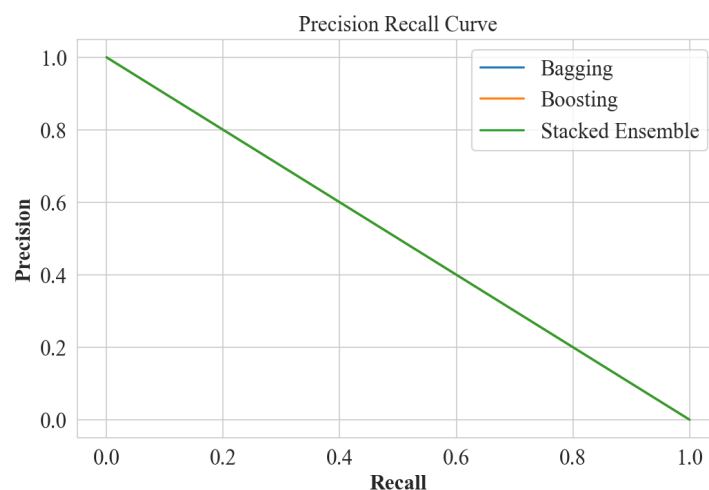


Figure 14: Precision Recall Curve

Inference: Boosting and Stacking maintain near-perfect precision across almost the full recall range, while Bagging's curve dips earlier, reflecting its comparatively higher false-negative rate on the malignant class.

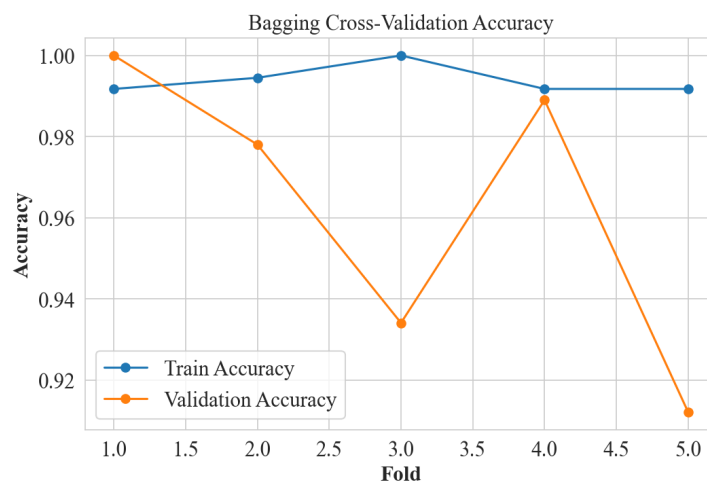


Figure 15: Bagging – Cross-Validation Accuracy

Inference: Bagging’s validation accuracy fluctuates noticeably across folds (from 91.2% to 100%) while training accuracy stays consistently near-perfect, indicating relatively higher variance across different data partitions compared to the other two ensembles.

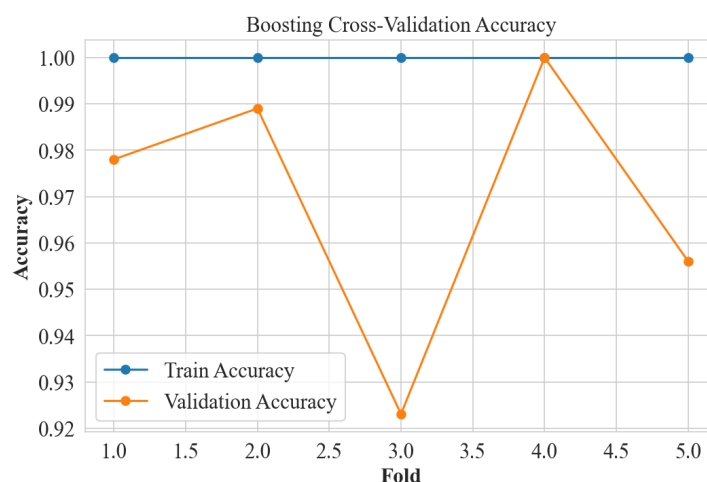


Figure 16: Boosting – Cross-Validation Accuracy

Inference: Boosting achieves perfect (100%) training accuracy on every fold, with validation accuracy ranging from 92.3% to 100% – the small but present train-validation gap is expected given AdaBoost’s tendency to fit training data very closely.

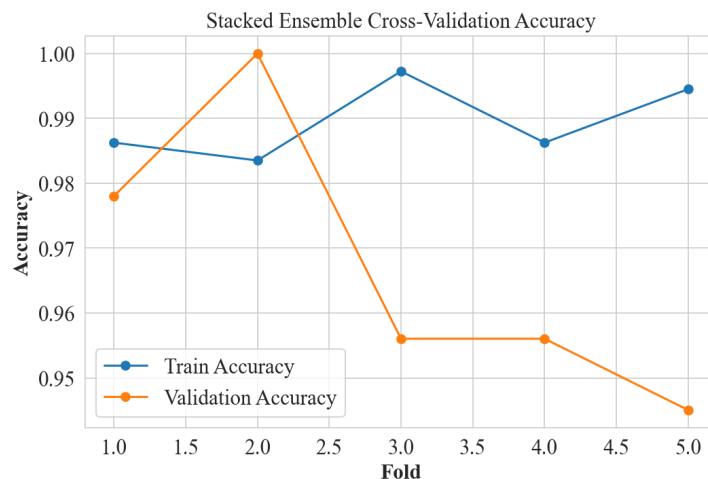


Figure 17: Stacked Ensemble – Cross-Validation Accuracy

Inference: The Stacked Ensemble shows the smallest train-validation gap of the three models (training consistently around 98-99%, validation 94.5-100%), suggesting its heterogeneous base learners and cross-validated meta-training combine to generalize slightly more stably than either Bagging or plain Boosting.

11 Bias–Variance Analysis

- **Bagging reduces variance:** By training 10 decision trees on independent bootstrap samples (`max_samples=0.8`) and averaging their votes, Bagging smooths out the high variance of individual unpruned decision trees; however, its cross-validation fold accuracies still swing the most (91.2%-100%) among the three models, indicating variance reduction here was only partial.
- **Boosting addresses bias:** AdaBoost’s sequential reweighting of misclassified samples directly targets the errors of previous weak learners, driving training accuracy to a perfect 100% on every fold and pushing test accuracy above both Bagging and Gradient Boosting – a clear demonstration of bias reduction through iterative correction.
- **Stacking benefits from heterogeneous models:** Because SVM, Naive Bayes, and Decision Tree make different, largely uncorrelated types of errors (margin-based vs probabilistic vs axis-aligned splits), their combination gives the Logistic Regression meta-learner complementary signals to exploit – though here, dropping Naive Bayes (leaving SVM+DT) actually improved performance, showing that diversity only helps when the added learner’s errors are genuinely informative rather than just different.
- **Best-performing method:** Boosting (AdaBoost) and the Stacked Ensemble tied for the best accuracy (97.37%) and confusion-matrix profile, with Stacking achieving a marginally higher ROC-AUC (0.9974 vs 0.9937), making Stacking the best overall performer when ranking quality is considered alongside raw accuracy.

12 Discussion

All three ensemble strategies performed strongly on this well-separated, moderately-sized dataset (569 samples, 30 features), with accuracies ranging from 95.6% (Bagging) to 97.37% (Boosting and Stacking). Boosting's bias-reduction mechanism and Stacking's heterogeneous-model combination both outperformed Bagging's pure variance-reduction approach, suggesting that on this dataset the dominant source of error for a single base learner was bias rather than variance. The Stacked Ensemble's slightly better ROC-AUC and smaller CV train-validation gap indicate it generalizes marginally more reliably, at the cost of roughly $2\times$ the training time of Boosting.

13 Conclusion

The Stacked Ensemble (SVM + Decision Tree base models with a Logistic Regression meta-learner) and Boosting (AdaBoost, `n_estimators=50`, `learning_rate=1.0`) achieved the best and statistically tied classification performance (97.37% accuracy) on the Wisconsin Diagnostic Breast Cancer dataset, both outperforming Bagging (95.61%). Given its marginally higher ROC-AUC and more stable cross-validation behavior, the Stacked Ensemble is the recommended model when maximum reliability is the priority, while AdaBoost remains an excellent, computationally cheaper alternative.

14 References

- [1] Scikit-learn: Ensemble Methods, <https://scikit-learn.org/stable/modules/ensemble.html>.
- Scikit-learn: Bagging Classifier, <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.BaggingClassifier.html>.
- Wisconsin Diagnostic Breast Cancer Dataset, UCI ML Repository, <https://archive.ics.uci.edu>.